https://doi.org/10.1038/s43587-023-00540-9

Long-term care insurance and the future of healthy aging in China

Simiao Chen, Linye Li, Lirui Jiao & Chen Wang

Check for updates

China faces urgent challenges associated with population aging. In this Comment, we summarize China's adoption of long-term care insurance and underscore its importance for social and economic wellbeing. We provide recommendations for a future of sustainable and healthy aging in China.

The World Health Organization (WHO) has emphasized the need for countries to have integrated long-term care (LTC) systems in the twenty-first century. LTC encompasses rehabilitative, preventative, assistive, palliative and social care, particularly emphasizing health management for geriatric individuals and those with chronic conditions[1]. Long-term care insurance (LTCI) offers financial support for essential care, including life care and basic nursing care, to older adults and those incapacitated by chronic illness. China — home to one of the world's largest and most rapidly aging populations — has initiated LTCI and bolstered its LTC system to enhance healthy aging outcomes. However, unique challenges persist. This Comment summarizes China's adoption of LTCI, underlines the social and economic importance of LTCI, and outlines recommendations to help China to achieve a future of healthy aging.

China's developing LTCI system for a future of healthy aging

In China, LTCI has substantially augmented the established social insurance system, which encompasses pension, medical, work-related injury and unemployment insurance[2]. The country has been proactive in rolling out LTCI. Figure 1 shows pivotal developments in this process, such as the establishment of an LTCI pilot program in 15 cities in 2016; the extension of the program to 20 further cities in two key pilot provinces (Shandong and Jilin); and the addition of another 14 pilot cities and areas (including autonomous prefectures and a district) by 2020. The pilot program has addressed LTC needs in the covered cities and areas, including through the provision of both life care and healthcare for older adults and people with disabilities. The pilots have covered home-based, community-based, and institutional care, with funding sourced primarily from medical insurance via the reallocation of pooled and personal resources. By 2022, the LTCI pilot program had attracted nearly 170 million participants across all pilot cities and areas, garnered 24.08 billion yuan in premium income against 10.44 billion yuan in expenses, and employed 331,000 LTC professionals in 7,679 facilities[3]. Other countries, such as Germany, Japan, and South Korea, have also established LTCI systems to address population aging. Despite China's progress, it trails these countries in generating legislative support for LTCI, prioritizing home- and community-based care, establishing state subsidization of LTCI, and implementing other important practices. These discrepancies highlight the challenges that remain for China in developing an effective national LTCI system. Below, we outline some of these challenges.

Inadequate investment. Despite China's recognition of the importance of LTCI, its financial commitment remains tentative. Many commentators and policymakers view LTCI predominantly as a financial burden rather than a strategic investment and overlook its profound social and economic benefits. According to OECD (Organisation for Economic Co-operation and Development) data, the average LTC expenditure of 17 member countries stood at 1.47% of gross domestic product (GDP) in 2018, and this share is projected to triple by 2050 (ref. 4). By contrast, China spent only 0.26% of its GDP on LTC in 2015 (ref. 5).

Vulnerable funding. China's primary financial support mechanism for LTCI, rooted in medical insurance funds across pilot cities and areas, is precarious. As in many countries implementing social LTCI, China's LTCI pilots follow a pay-as-you-go approach. However, China's LTCI pilots rely predominantly on social medical insurance programs for premium collections. By contrast, most other countries with LTCI have distinct funding streams. Data from the National Health Security Administration indicate that as of June 2018, nearly 80% of all LTCI premium income across 15 pilot cities came from social medical insurance accounts, 12% came from individual participants and 9% came from public finances[6]. This heavy reliance could strain China's medical insurance programs, thereby risking their sustainability. The lack of a dedicated funding avenue threatens the long-term viability of LTCI, which will potentially impede China from realizing the comprehensive social and economic benefits that LTCI is poised to offer.

Uneven access. There are clear disparities in LTCI accessibility in China. A breakdown of pilot cities and areas based on enrollee categories shows that 15 cities (and areas) exclusively accommodate 'Urban Employees Basic Medical Insurance' (UEBMI) recipients at present. Meanwhile, 13 cities (and areas) that initially covered only UEBMI recipients are in the process of broadening their reach to 'Urban and Rural Residents Basic Medical Insurance' (URRBMI) enrollees, and 21 cities (and areas) cater to both groups. This uneven and inequitable approach to LTCI implementation — in which urban residents are favored over rural residents — contrasts with the universal models used in countries such as Germany, Japan, and South Korea. As the average health status of rural older people is worse than that of their urban counterparts, China's current strategy disproportionately advantages urban residents and exacerbates social divides. Marginalized groups (especially rural inhabitants) are left grappling with limited LTC access, which results in the amplification rather than attenuation of existing social security gaps.

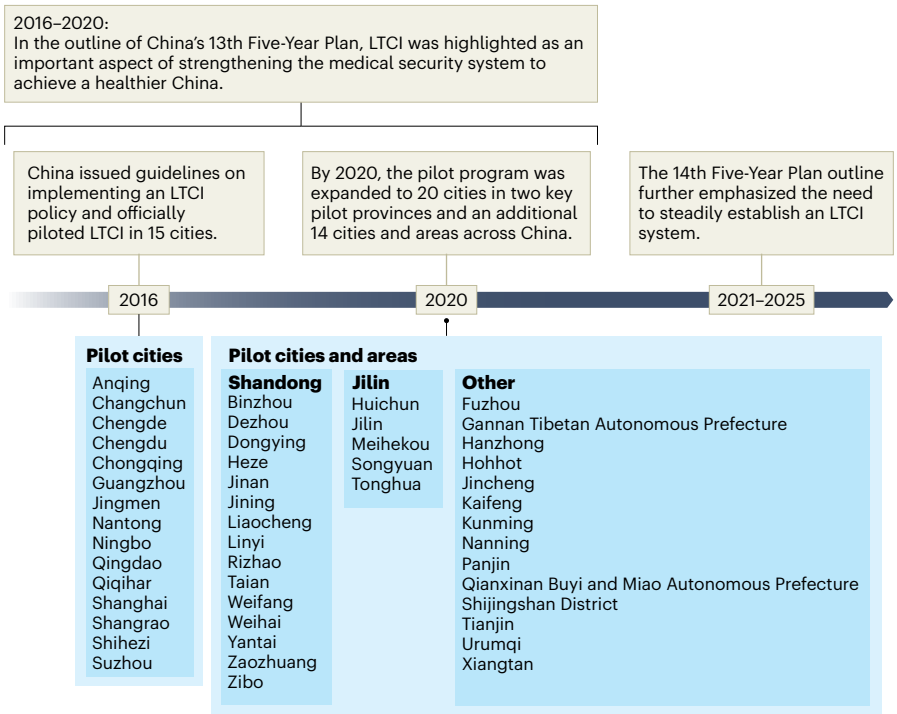

Fig. 1 | Timeline of recent events relevant to LTCI in China. This is a timeline of recent events relevant to LTCI in China[15], including the establishment of LTCI pilot programs in 15 cities in 2016 as well as the program's extension to 20 cities in two key pilot provinces (Shandong and Jilin) and another additional 14 pilot cities and areas by 2020. 'Other' includes cities, autonomous prefectures, and a district.

Broader social and economic importance of establishing LTCI in China

Although the merits of LTCI for reducing disease burden and promoting population health are widely recognized[7], its broader social and economic effects deserve greater exploration. Below, we propose several likely socioeconomic outcomes of investing in LTCI.

Nurture a productive workforce. By addressing the needs of China's older adults and people with disabilities, LTCI reduces the care burden placed on younger family members, which might otherwise force them to leave their jobs. Indeed, owing to common cultural beliefs that adult children have a responsibility to provide care for their older parents (which stands to put substantial strain on a generation of adults (and in particular, women) born under the One Child Policy), a shift toward care performed by LTC professionals would enable these family members to remain within the labor force and promote gender equality. In addition, the implementation of LTCI alleviates personal financial pressures (for example, through reducing costs related to activity of daily living[8]) and associated stress, thereby fostering an environment that is conducive to increased productivity. Moreover, LTCI can enhance the efficiency of healthcare utilization (for example, tertiary hospitals reduced inpatient stays by 41.0% following the implementation of LTCI[9]) and reallocate healthcare resources to others more in need, thereby improving health outcomes and increasing productivity.

Bolster the healthcare industry. The growing need for LTC heralds a wealth of employment opportunities that span from hands-on caregiving roles such as nursing to support roles in administration. The expanding LTC landscape will stimulate businesses to innovate and expand their range of healthcare solutions. This strategic shift could position China not only as a primary consumer, but also as an international exporter, of cutting-edge healthcare solutions. A sophisticated LTCI system can further catalyze investment in health-related research and development (R&D), potentially yielding breakthroughs in geriatric care and advanced medical technologies. As a relevant historical comparison, one study discovered that from 1965 to 2005, the expansion of health insurance propelled the growth rate of medical R&D by an impressive 57%[10]. Another study that investigated the mechanisms behind this growth revealed that insurance expansion not only amplifies medical innovation by enlarging the market but also sharpens physician efficiency in contributing to medical R&D, partly because a more favorable ratio of insured individuals per physician and more generous payment rates encourage physicians to adopt and innovate new technologies[11].

Stimulate the healthcare economy. The future of the economy is intimately connected with the future of healthcare, from both the demand side and the supply side. Healthcare demand has increased with the rapidly growing aging population in China and investing in LTCI is an important government response to address this rising demand[12]. With respect to the supply side, the 'Healthy China 2030' blueprint forecasts that by 2030, China's healthcare market size is expected to reach 16 trillion yuan, which will make it a pillar of China's economy. These trends are likely to ripple beyond the healthcare industry and fuel sustained and comprehensive economic development. In the current scenario (in which the healthcare supply falls short of meeting the burgeoning

demand in China), strategic initiatives such as investing in LTCI are crucial for steering the country towards a health-centric society.

Recommendations and prospects for LTCI in China

The Lancet Commission on Healthy Aging in China has made several recommendations that are relevant to LTCI and that include (1) unifying regionally segmented LTCI initiatives and striving for a national scheme that is independently funded; (2) subsidizing consumers of age-related care rather than providers to facilitate market competition; (3) considering the introduction of an individual savings component to the national LTCI financing strategy; and (4) introducing portable LTCI accounts (for example, allowing migrant adults with household registrations outside of the pilot cities and areas to maintain and benefit from LTCI accounts) or savings accounts designated for future purchase of LTCI (to attract the participation of migrant adults who are currently of working age)[6].

In addition to adopting the Lancet Commission's proposals, we suggest that China should also consider implementing the following recommendations with respect to LTCI.

Repositioning LTCI as a strategic investment. Instead of perceiving LTCI as a mere expense, China should view it as a strategic investment in the country's health and future. The WHO emphasizes that countries should mobilize resources, both locally and globally, to prioritize healthcare as a vital industry, and place population health at the center of economic activities. Recent studies have revealed that many major economies underinvest in health, with China being a prime example[13]. Under these conditions, China should see increased investment in healthcare as not only a necessary corrective to past underinvestment, but also as a means of developing a dynamic economy for the future.

Universal access and a diverse LTC ecosystem. The gap between urban and rural healthcare access can be bridged through centralized funding of LTCI. Providing steady funding and narrowing this gap is crucial for rural areas, which often lack adequate healthcare resources. Furthermore, by engaging a variety of stakeholders, from private corporations to charitable nongovernmental organizations, China can create a diverse LTC landscape. Such an inclusive approach will ensure that the varied needs of its older adults and citizens with disabilities are met.

Institutionalize oversight mechanisms. Empirical evaluations of the effectiveness of the LTCI pilot policy have found that LTCI implementation has several positive effects on older adults' healthcare utilization and finances[7]. However, currently China's LTCI approach lacks an evidence-based feedback loop for periodic adjustment. We propose that a consistent monitoring and evaluation mechanism is vital to ensure the effective use of funds and to prevent potential mishandling. Such evaluation should incorporate substantial sample sizes to reflect all cities and areas included in the scheme, use the full timescale of the LTCI rollout, and employ rigorous analytical methods. Reviewing how other countries approach this issue might be instructive. For example, in Germany, following the Long-term Care Further Development Act that was passed in 2008, the Medical Review Board initiated regular quality inspections of both home care and nursing home care providers[14]. The Medical Review Board must also publish the inspection results to meet transparency requirements. Furthermore, LTCI fund administrators and care providers have collectively constructed an accessible monitoring system in which inspection results are graded in a standardized fashion. It is critical for China to integrate insights from such oversight practices, as well as scholarly research, to ensure the transparency and accountability of LTCI.

Conclusion

As China grapples with the challenges brought about by imminent population aging, proactive measures are paramount. The promotion and facilitation of formal LTC through LTCI stands as a potential cornerstone for strengthening the health system. LTCI promises not only improved health outcomes for older adults and people with disabilities but also the cultivation of a vibrant labor force, an invigorated healthcare industry, and a buoyant healthcare economy. Embracing LTCI underscores a society's respect for its older adults and citizens with disabilities and lays the groundwork for healthy aging and longevity equity through a sustainable person-centered approach. As China progresses in its LTCI endeavors, it should recognize the positive social and economic effects of LTCI, advocate for its strategic value, ensure universal access, foster a diverse LTC landscape, and buttress these efforts with robust oversight.

Simiao Chen[1,2] ✉, **Linye Li**[3], **Lirui Jiao**[4] **& Chen Wang**[2,5,6]

[1]Heidelberg Institute of Global Health (HIGH), Faculty of Medicine and University Hospital, Heidelberg University, Heidelberg, Germany. [2]School of Population Medicine and Public Health, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China. [3]Chinese Academy of Social Sciences, Beijing, China. [4]Department of Health Policy and Management, Gillings School of Global Public Health, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA. [5]National Clinical Research Center for Respiratory Diseases, Beijing, China. [6]Department of Pulmonary and Critical Care Medicine, Center of Respiratory Medicine, China-Japan Friendship Hospital, Beijing, China.

✉e-mail: simiao.chen@uni-heidelberg.de

Published online: 13 December 2023

References

1. World Health Organization. Integrated Continuum of Long-term care. who.int, https://www.who.int/teams/maternal-newborn-child-adolescent-health-and-ageing/ageing-and-health/integrated-continuum-of-long-term-care (accessed December 9, 2023).
2. Feng, Z. et al. *Lancet* **396**, 1362–1372 (2020).
3. National Healthcare Security Administration. Statistical Communiqué on the Development of National Healthcare security [in Chinese]. *nhsa.gov.cn*, http://www.nhsa.gov.cn/art/2023/7/10/art_7_10995.html (2023).
4. Lee, S.-H., Chon, Y. & Kim, Y.-Y. *Health Care* **11**, 206 (2023).
5. Li, F. & Otani, J. *Int. J. Health Plann. Manage.* **33**, 479–488 (2018).
6. Chen, X. et al. *Lancet* **400**, 1967–2006 (2022).
7. Chen, S. et al. *J. Glob. Health Econ. Policy* **1**, e2021014 (2021).
8. Lei, X., Bai, C., Hong, J. & Liu, H. *Soc. Sci. Med.* **296**, 114745 (2022).
9. Feng, J., Wang, Z. & Yu, Y. *Soc. Sci. Med.* **258**, 113081 (2020).
10. Frankovic, I. & Kuhn, M. *J. Health Econ.* **87**, 102717 (2023).
11. Clemens, J. *The Effect of US Health Insurance Expansions on Medical Innovation* (National Bureau of Economic Research, 2013).
12. Zhu, Y. & Österle, A. *Soc. Sci. Med.* **191**, 30–37 (2017).
13. Chen, S., Kuhn, M., Prettner, K., Bloom, D. E. & Wang, C. *Soc. Sci. Med.* **287**, 114270 (2021).
14. Rothgang, H. *Soc. Policy Adm.* **44**, 436–460 (2010).
15. National Healthcare Security Administration, Ministry of Finance of the People's Republic of China. Guiding opinions on expanding the long-term care insurance pilots [in Chinese]. *nhsa.gov.cn*, http://www.nhsa.gov.cn/art/2020/9/16/art_37_3586.html (2020).

Acknowledgements

The authors of this Comment received funding from the Chinese Academy of Engineering (project 2022-XBZD-14), the Chinese Academy of Medical Sciences & Peking Union Medical

College (project 2022-CFT-QT-016), Horizon Europe (HORIZON-MSCA-2021-SE-01) (project 101086139 — PoPMeD-SuSDeV), and the Bill & Melinda Gates Foundation (project INV-006261).

Author contributions

S.C. and C.W. contributed to the conceptualization of this Comment. S.C., L.L., L.J. and C.W. contributed to the writing of this Comment.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Aging* thanks Yanan Zhang for their contribution to the peer review of this work.